# Bihar Engineering University, Patna

## B.Tech. 1st Semester Examination, 2023

*Course: B.Tech.* *Time: 03 Hours*

*Code: 102101* *Subject: Physics (Electromagnetism)* *Full Marks: 70*

***Instructions:-***

*(i) The marks are indicated in the right-hand margin.*
*(ii) There are **NINE** questions in this paper.*
*(iii) Attempt **FIVE** questions in all.*
*(iv) Question No. 1 is compulsory.*
*(v) Symbols used (if any) have their usual meanings*

**Q1** ***Choose the correct answer of the following (Any seven question only):*** **[2 x 7 = 14]**

(a) The curl of electric field is
(i) always zero
(ii) never zero
(iii) sometimes zero
(iv) undefined

(b) Faraday cage protects from lightning strike because the cage material is
(i) very strong
(ii) conducting so it grounds the electricity
(iii) conducting so it shields electromagnetic waves
(iv) none of the above

(c) Magnetic monopoles do not exist because
(i) they are extremely difficult to find
(ii) the magnetic field is solenoidal
(iii) monopoles become stable at extremely low temperatures
(iv) electric field is much stronger than magnetic field

(d) Ampere's law is incomplete because
(i) it was discovered many years ago
(ii) it is incompatible with the law of conservation of charge
(iii) it is valid only for time dependent currents
(iv) it is incompatible with the solenoidal nature of magnetic field

(e) Which of the following is true
(i) Maxwell's equations are valid only for electrostatics
(ii) The divergence of magnetic field is zero only in magnetostatics
(iii) Maxwell's equations hold only in empty space
(iv) Maxwell's equations are valid under all circumstances

(f) For a constant magnetic field $\vec{B} = B_0 \hat{k}$ the vector potential $\vec{A}$ is
(i) 0
(ii) $B_0(-y\,\hat{\imath} + x\,\hat{\jmath})/2$
(iii) $B_0(y\,\hat{\imath} + x\,\hat{\jmath})/2$
(iv) $B_0(x\,\hat{\imath} + y\,\hat{\jmath})/2$

(g) If you put a Dielectric slab between the plates of a parallel plate capacitor
(i) The electric field inside increases
(ii) The capacitance increases
(iii) The capacitance decreases
(iv) The potential difference between plates increases

(h) Magnetic susceptibility of diamagnetic material is
(i) Positive (ii) Negative (iii) Zero (iv) None of above

(i) Lenz's law is consequence of the law of conservation of
(i) Charge (ii) Energy (iii) Momentum (iv) Mass



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(j) Which of the following is not true about the electromagnetic waves
(i) They are transverse in nature
(ii) Electric field is perpendicular to magnetic field
(iii) Electric field is parallel to propagation vector **K**
(iv) Magnetic field is perpendicular to propagation vector **K**

**Q.2** (a) Show that (i) electrostatic field is always normal to the surface of a conductor and (ii) electrostatic potential is always constant inside conductor [6]

(b) If $\vec{E} = a(y\hat{\imath} - x\hat{\jmath})$ show if this electrostatic field can exist or not. [4]

(c) The electric field E in the x-y plane is given by $\vec{E} = 2cx\hat{\imath} + ay\hat{\jmath}$ , where c and a are constant, what is the charge density responsible for this field [4]

**Q.3** (a) A spherical conductor contain a uniform surface charge density σ determine the field and potential due to charge distribution. [7]

(b) State Gauss's law of electrostatics. Derive differential form of Gauss's law [7]

**Q.4** (a) Obtain detailed boundary conditions on the electric field and electric displacement. [6]

(b) What is electric dipole? Define its dipole moment [2]

(c) Find expression for the field and potential due to electric dipole [6]

**Q.5** (a) Write brief technical notes on ferromagnetic, paramagnetic and diamagnetic material. [8]

(b) What is magnetic vector potential? A current distribution gives rise to magnetic vector potential, [6]

$$\vec{A}(x, y, z) = x^2 y\hat{\imath} + y^2 x\hat{\jmath} - xyz\hat{k},$$

find the magnetic field at ( -1,2,5)

**Q.6** (a) Derive the expression for the energy stored in a magnetic field [6]

(b) Starting from the Faraday's law obtain its differential form. Establish the equivalence of Faraday's law and motional emf. [4]

(c) State and discuss Lenz's law. [4]

**Q.7** (a) Obtain the continuity equation for charge and use it to modify Ampere's law to include Displacement current. [6]

(b) State and derive Poynting's theorem. [8]

**Q.8** (a) Derive the electromagnetic wave equation in vacuum. Prove the transverse nature of plane wave. [6]

(b) For a plane wave given as $\vec{E}(\vec{x}, t) = A\sin(\vec{k}.\vec{x} - \omega t)\hat{\jmath}$ find (i) the magnetic field (ii) direction of propagation (iii) Poynting vector and (iv) the energy density. [8]

**Q.9** (a) Starting with general Maxwell's equation derive Maxwell's equation in a linear medium with permittivity $\epsilon$ and permeability $\mu$. [8]

(b) Write a technical note on the Method of Images. Mention the significance of uniqueness theorem as a basis for the method. [6]

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